

Multiple Choice

1. **Answer:** A

Options: A) 102.5; B) 100; C) 109.5; D) 106.78; E) 101.22 *Note:* Correct option: A - 102.5

1. **Answer:** A

Options: A) nitrogen; B) a halogenated hydrocarbon; C) chlorine; D) copper; E) arsenic

Note: Correct option: A - nitrogen

2. **Answer:** A

Options: A) NaCl; B) LiCl; C) AlCl₃; D) KCl; E) FeCl₃

Note: Correct option: A - NaCl

3. **Answer:** A

Options: A) Boiling point is 101.00; Freezing point is 1.55; B) Boiling point is 101.00; Freezing point is -2.00; C) Boiling point is 99.57; Freezing point is 1.55; D) Boiling point is 100.0; Freezing point is 0.0; E) Boiling point is 101.00; Freezing point is 0.0

3. **Answer:** E

Options: A) 10.0 g; B) 24.4 g; C) 15.2 g; D) 22.3 g; E) 19.6 g

Note: Correct option: E - 19.6 g

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'AlBr₃'.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '4'.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '1.98 cal mol⁻¹K⁻¹'.

Long Form Response

11. **Answer:** .9342 V. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.
12. **Answer:** 0.0100 mol. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.